PHOTOSYNTHESIS

29/08/22

Photosynthesis:-
- the process by which plants synthesize carbohydrates from raw materials using energy from light.

word equation:-

carbon dioxide + water → glucose + oxygen

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$

in the presence of light and Chlorophyll

chlorophyll:-
- a green pigment that is found in chloroplasts.

* it transfers light energy to chemical energy for photosynthesis

the use and storage of the carbohydrates produced:-

* (glucose makes cellulose, starch, sucrose and nectar)

starch → energy store
cellulose → build cell walls
Glucose → respiration (to provide energy)
sucrose → transport in phloem
nectar → attract insects for pollination

* nitrate ions make amino acids
* magnesium ions make chlorophyll

30/08/22

Investigate the need for chlorophyll, light and carbon dioxide for photosynthesis, using appropriate controls :-

- chlorophyll is important because it helps absorb the 'light' needed for photosynthesis
- CO_2 is important because it is converted into sugars
- light is important because it acts like the fuel or energy to drive the reaction.

* chlorophyll provides the site for photosynthesis, this is why organisms without chloroplast cannot photosynthesize.
* controls → temperature and oxygen (levels)

Investigate the effects of varying light intensity, CO_2 concentration and temperature on the rate of photosynthesis :-

- without enough light, a plant cannot photosynthesize very quickly even if their is plenty of water and CO_2 and a suitable temperature.
  - increasing the light intensity means increasing the rate of photosynthesis unless a limiting factor is lacking.
- if the concentration of CO_2 is increased then the rate of photosynthesis will increase too.
- as with other enzyme-controlled reactions, the rate of photosynthesis is affected by temperature.
  - at low temperatures, the rate of photosynthesis is limited by the number of molecolar collisions between enzymes and substrates.
  - at high temperatures, the enzymes are denatured.

31/08/22

Investigate and describe the effects of light and dark conditions on gas exchange in an aquatic plant (*using hydrogencarbonate indicator solution):-

* hydrogencarbonate indicator can detect increases and decreases in carbon dioxide concentration.
* normally is red, an increase and it turns yellow, and a decrease and it will turn purple.

- 3 tubes:
  - 1st = 10cm pond weed + tinfoil + hydrogencarbonate (indicator)
  - 2nd = 10cm pondweed + hydrogencarbonate indicator.
  - 3rd = CONTROL with hydrogencarbonate indicator only.

Leave test tubes for 2-3 hrs.

1st → yellow (CO_2 was added bc of respiration)
2nd → purple (CO_2 was taken for photosynthesis)
3rd → red (bc its control)

Limiting factors:-

Sunlight:
Light intensity increases → rate of photosynthesis increases until the plant is photosynthesizing as fast as it can, now even if the intensity increases, the plant photosynthesizes at the same rate.

CO_2:
amount of CO_2 increases → rate of photosynthesis increases until the maximum is reached.

temperature:
chemical reactions are working very slowly in low temperature so a plant photosynthesizes faster on a warm day than on a cold day.

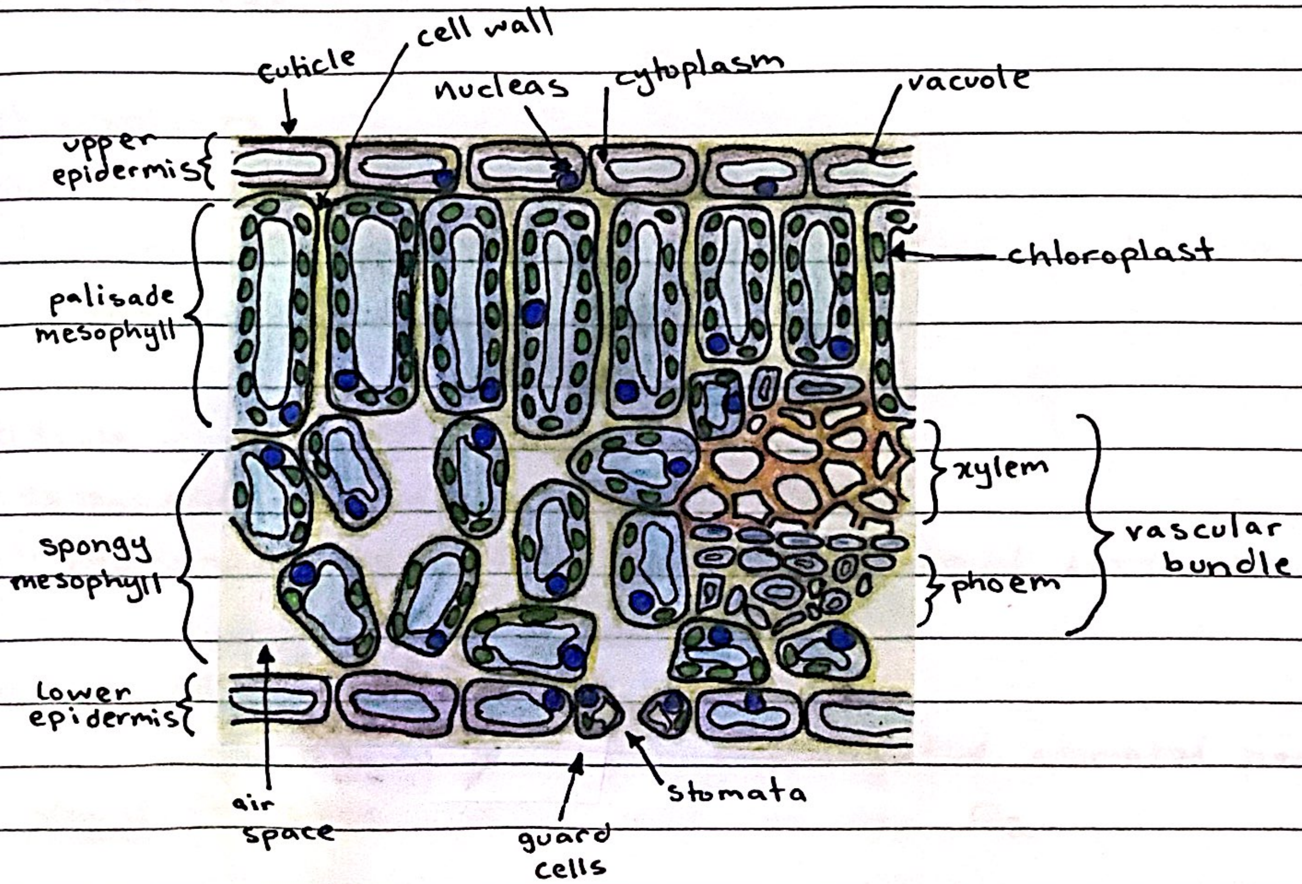
LEAF STRUCTURE

31/08/22

* most leaves have a <u>large surface area</u> and are <u>thin</u>.

large surface area → to absorb more light
thin → short distance for CO_2 to diffuse into cells.

- the large surface area means that their are more chloroplasts, also more sunlight is being absorbed, and the number of stomata is more which plays a crucial role in gaseous exchange.



Leaf Structure

01/09/22

epidermis:-

- thin and transparent
  - allows more light to reach the palisade cells

palisade mesophyll:-

- at the top of leaf with many chloroplasts
  - absorbs all available light and increases the rate of photosynthesis.

spongy mesophyll:-

- site of gaseous exchange
  - air spaces to allow gases to diffuse through the leaf.

chloroplasts:-

- contain chlorophyll
  - more in number on the upper surface as there is more light exposure their.

cuticle:-

- extracellular hydrophobic layer
  - protects against water loss and external stresses.

guard cell:-

- controls gas diffusion by opening/closing the stomatal pores.
- stomata → takes in CO_2 and releases O_2.

vascular bundle:-

- transports critical substances to various parts of the plant.
- xylem transports water and nutrients.
- phloem transports organic molecules
* cambium is involved in plant growth